Natural resources and their uses

1. Key Points & Definition of Resources

- **Definition:** Anything provided by nature that has value and utility can be termed as a **'Resource'**.
- **Conditions for Utility:** A resource only becomes useful when it is accessible, usable, and profitable to obtain. Cultural acceptance also plays a vital role (e.g., Coal).
- **Role of Technology:** Technology plays a huge role in making resources accessible and usable.

Key Requirements for a Resource

RESOURCE

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Culturally Accepted | Usable & Accessible | Profitable

2. Classification of Resources

- **By Use:** Essential for life, Building & Ornamental, Fuels & Energy.
- **By Availability:**
  - **Renewable:** Restored and regenerated naturally on a continuous basis (e.g., sunlight).
  - **Non-Renewable:** Takes millions of years to form naturally (e.g., coal).
- **By Source:** Human-made and Natural.
- **Human Resources:** Humans are also a resource; this includes the knowledge, skills, and abilities of a person.

> **International Solar Alliance (IASE) — India's Leadership in Renewable Energy:**
> India and France launched the **International Alliance for Solar Energy (IASE)** in 2015 — a coalition of sunshine-rich countries committed to harnessing solar power.
> - Focuses on countries blessed with abundant sunlight throughout the year.
> - India has helped channel billions of dollars into solar projects.
> - The **Bhadla Solar Park** is a symbol of India's solar ambitions.

3. Regional Case Studies & Observations

- **Uneven Distribution:** Resources are divided on Earth unevenly. This causes the establishment of human societies, trades, and sometimes conflicts/wars.

- **Punjab (Groundwater Depletion):** Known as the *'Granary of India'* due to agricultural development from the Green Revolution. However, regular and heavy extraction has made several areas groundwater-deficient.
- **Kaveri River Dispute:** An important river in South India sustaining lives and historically important since the Cholas. States through which it flows (Karnataka, Tamil Nadu, Kerala) dispute water allocation, highlighting that fair resource division and management are critical.
- **Sikkim (Harmonious Co-existence):** A prime example of how modernity and harmony with nature can co-exist. Known for organic farming, farmers use natural fertilizers and manure, yielding higher market prices for organic goods while preserving the environment.

4. Resource Extraction & Industrial Impact: Cement Production

- **Importance:** Cement is a vital human-made resource required for modern infrastructure, buildings, and development.
- **Environmental & Resource Losses:**
  - **Carbon Emissions:** Cement manufacturing is a major source of global greenhouse gas emissions, directly contributing to global warming.
  - **Depletion & Land Loss:** Rapid extraction of natural limestone reserves and large-scale land/topsoil degradation due to quarrying and mining.

Problems of Excessive Use of Natural Resources

Excessive Use of Natural Resources

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- Climate Change
- Global Warming
- Pollution
- Biodiversity Loss
- Less Resources for Future

Conclusion: This is the reason why we should use resources judiciously and sustainably to preserve them for future generations.